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RFID Based Real-time Recognition of ongoing Gesture with Adversarial Learning

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ABSTRACT

At present, wireless sensing based gesture recognition is becoming a rising star due to its convenience and non-invasiveness without privacy issues, while the strict requirement of the deployment and surrounding environment is still an unavoidable issue which limits its development and generalization. Although there are some works involving the environmental variance, the changes of relative positions between devices and users are ignored. As one of the most popular wireless sensing methods, RFID is widely used in activity recognition with its stable low-level physical characters such as phase and RSS. Besides, the signals reflected from RFID tags intuitively delineate its movements. On the other hand, many interactive gesture-driven applications, such as gesture input for video games, have a paramount and unavoidable issue about the latency between completion of a gesture and its recognition. Inspired by deep learning, this paper presents a real-time ongoing gesture recognition system EUIGR, which efficiently integrates phase and RSS data streams, and extracts both environment and user invariant features. The proposed system seamlessly integrates CNNs (Convolutional Neural Networks) and LSTM (Long Short-Term Memory) to fuse RFID low-level physical characters and extract space-temporal information. Furthermore, with adversarial learning, EUIGR suppresses environment-related factors and the user-specific features, and obtains strong robustness to individual diversity and decreases the environmental dependence. We also implement the system with COTS RFID devices, and extensive experimental results show the effectiveness and accuracy of EUIGR.

CCS CONCEPTS

• **Human-centered computing** → *Gestural input*; *Ubiquitous and mobile computing systems and tools.*

*Dong Wang is the corresponding author.

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KEYWORDS

RFID, wireless sensing, ongoing gesture recognition, adversarial learning

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1 INTRODUCTION

As an an important method of Human-computer interaction (HCI) technology, human gesture recognition builds a friendly and straightforward bridge between human and computer compared with the text-based and graphic user interface, which draws great attention in recent years with the rapid development of the artificial intelligence, mobile computing, Internet of Things, virtual reality, argument reality and other technologies[1, 22, 29, 35, 39].

The state-of-the-art works on gesture recognition have used a range of different sensor data, including video from the camera, inertial data from the wearable sensors, signal stream from the wireless wave, etc. However, the method based on computer vision faces the risk of disclosure of privacy, while wearable sensors require batteries. In recent years, wireless sensing is a hot topic in both academic and industry, while the strict requirement of the deployment and surrounding environment is still an unavoidable issue which limits its development and generalization. Although there are some works involving the environmental variance, the changes of relative positions between devices and users are ignored. Passive RFID (Radio Frequency Identification) is widely employed in an ocean of real-world applications[8, 30, 38, 40, 43] due to low-cost, mini-size and battery-free characters[13]. It has also achieved great success in gesture recognition[6, 18, 29, 44] and indoor localization[25, 34, 37] using RSS or phase information reported by COTS (Commercial Off-The-Shelf) RFID readers. Although exhibiting great potentials, RFID based gesture recognition methods can handle gestures due to dynamic environments and degrade significantly when it comes to user diversity and environment variety[18, 44]. In addition, a number of recent works on gesture detection are designed for recognizing gesture after it is completed[9, 22], so the latency between gestures completion and recognition cannot be avoided.

In this paper, we explore the possibility of leveraging RFID to recognize ongoing gestures in a convenient and robust way. Our purpose is fusing multimodal RFID data and extracting space-temporal information to enable a general, pervasive, environment-independent, user-invariant and real-time gesture-driven interactive system, there are several challenges that need to be addressed:

- **How to recognize gesture before it is completed:** The user experience about real-time gestures-driven interactive applications is highly related to the latency. Users want the gestures can be recognized when they begin acting it if possible and cannot afford on the lag caused by the execution of the recognizing step which begins at the end of the gesture.
- **How to fuse multimodal RFID data:** Multimodal sensor data fusion is a common challenge in the activity recognition field. The phase and RSS carry the information from different aspects, and when extracting the features from them manually and respectively, the information which is implicit in the relation between them is not used effectively and is unable to generate the best representations of the gesture.
- **How to extract environment & user invariant features:** The information about the gestures carried by the wireless signals is specific to the user who executes the gestures, i.e, height, weight, body shape. And the signal propagation path is determined by the relative position of the RFID tag and the antenna, as well as the surrounding environment. All these factors may affect the signal in different ways. Accordingly, the gesture recognition model trained for some users may work well with these users in a certain environment, while it has poor performance with other users who do not take participate in the training process and degrades more seriously when it comes to the environmental variety and even the change of relative positions.

To address the above challenges, this paper proposes EUIGR, an RFID based Environment independent and User Invariant real-time Gesture Recognizing system, which focuses on the latency of recognition, the diversity across environments, the variety among users, and the fusion of Multimodal RFID data. The key contributions of this paper are as follows:

Low latency and high accuracy: To achieve high recognition accuracy, we build a multimodal architecture for RFID data, which processes phase and RSS information using CNN respectively, instead of extracting and fusing features manually. Then unlike recent works [3, 4, 44] which combine segmentation and classification stages, extract the whole gesture and then classify it, we develop an LSTM based sequence labeling classifier, where the probability of each gesture is given at every time clip. Consequently, EUIGR achieves a negative delay, which means that a gesture is detected before it is completed.

Environment & user invariant training: With a multi-task optimization in environment & user invariant training (EUIT) via adversarial learning, we successfully suppress the effect of user-volatile and environment-specific factors in gesture representations. As a result, EUIGR achieves the average precision and recall of 96.6% and 96.0% respectively for gesture recognition with unseen users, and leads to the average 88.6% of precision and 86.7% of recall in untrained positions.

A wealth of experiments: At last, we evaluate our system in a wrist-worn-tag deployment, take eight traffic command gestures as experimental subjects. Extensive experiments demonstrate that EUIGR achieves an average recall of 96% even at the midpoint of a gesture with unseen users, and has strong robustness to individual diversity and environmental variety. T-distributed stochastic neighbor embedding (t-SNE)[23] visualization depicts that the deep representations distributions of different users are well aligned with each other, which demonstrates the strong capability of adversarial architecture in reducing domain-variability.

The rest of this paper is organized as follows. After reviewing related works in Section 2, we propose the background and briefly introduce the system components in Section 3. Furthermore, Section 4 illustrates the detailed design of the gesture recognition system with negative delay[14] and EUIT. In Section 5, the performance evaluations of EUIGR are analyzed. Section 6 conducts a discussion of this work and several thoughts on its future. Finally, Section 7 concludes this paper and discusses future work.

2 RELATED WORKS

Our work is inspired by gesture recognition, ongoing gesture detection, and domain adversarial learning.

2.1 Gesture Recognition

Existing gesture recognition related works can be classified into three categories.

Wearable sensor based: This line of work deals with the problem of gesture recognition with wearable sensors[12, 27, 28, 39]. At present, a smartwatch or smart wristband commonly integrates more and more sensors. For example, wrist-worn devices containing inertial sensors are utilized to recognize the eating gesture[28] and identify smoking gestures[27], translate sign language[39, 41], and identify fine-grained interactive gestures[12], respectively. All these methods require users to charge devices from time to time.

Camera based: The camera is applied to capture image sequences, then the computer vision techniques are applied to track the motion of a person. Most recently, researchers have widely exploited Leap Motion or Kinect to build gesture recognition systems to enable sign language translation at both word and sentence levels with high accuracy[9], to recognize continuous sign language with weakly supervised learning[5]. However, camera based methods face the risk of disclosure of privacy.

Wireless based: In recent years, wireless signals such as ultrasound, WiFi, RFID, mmWave are widely used for fine-grained interactive gestures recognition[19, 20, 36], sign language translation[22], keystrokes recognition[1], and limb-level gestures detection[3, 6, 29, 44]. Due to the environment sensitivity of wireless signal, many gesture recognition systems based on wireless signal work well in relatively limited and controlled environment and achieve high accuracy on some special users, but the performance may degrade in some unstable or uncontrolled environments. Because RFID signal is quite stable among different environments, and with domain adversarial learning, EUIGR make some progress in the robustness to user diversity and environment variety. Many previous works on gesture recognition usually take two stages including segmentation

and recognition, EUIGR implements a pipeline to deal with the raw signals and inferences gesture prediction in real-time.

2.2 Ongoing gesture detection

In recent years, some researchers find the problem of ongoing gesture recognition and try to achieve the zero or negative lag[14, 15, 33]. Tran et al. [33] propose a smart watch based early gesture detection technique, and achieve the precision of 87.4% at the midpoint of stroke gestures of the table-tennis game. Molchanov et al.[14] use deep learning techniques to predict twenty-five hand gestures online from videos. Hoai et al.[15] design computer vision based early event detectors which enable sign language translation, emotions recognition, respectively.

2.3 Domain Adversarial Learning

Adversarial learning are utilized to learn robust representations which are discriminative on source domain while invariant between domains[10, 32]. Zhao et al.[42] combine CNN and RNN with adversarial learning to extract sleep-specific and subject-invariant features from RF signals to predict the sleep stages. SIT [24] utilizes multi-task optimization to reduce variance across speakers and keep representations senone-discriminative. Sriram et al. [31] present a robust end-to-end speech recognition framework using generative adversarial network in a data-driven way. Unlike them, EUIGR takes the ongoing gesture into consideration and the training objectives of the user domain, environment domain and the gesture domain undertake different tasks, namely classification and sequence labeling respectively.

3 SYSTEM MODEL

In this section, we present the RFID communication model of our system. Besides, we roughly introduce the overall of EUIGR, the system designed for the purpose of practical gesture recognition with low latency, high accuracy and strong robustness.

3.1 RFID Communication Model

Present COTS RFID readers[16] can report the low-level signal characters like RSS ($A(t)$) and phase ($\varphi(t)$) when interrogating a tag successfully. The received backscatter signal $S(t)$ can be presented by:

$$S(t) = A(t)e^{-j\varphi(t)} \tag{1}$$

where both $A(t)$ and $\varphi(t) = 4\pi d(t)/\lambda$ are dependent on the propagation distance $d(t)$. In practice, the signal $S(t)$ is the superposition of signals from the LOS (line of sight) path (only reflected by tags) and other indirect paths (reflected again by nearby obstacles), named as multipath. Therefor, with equation (1), it can be divided into two parts as:

$$\begin{aligned} S(t) &= S_{los}(t) + S_{mul}(t) \\ &= A_{los}(t)e^{-j\varphi_{los}(t)} + \sum_{i}^{NLOS} A_i(t)e^{-j\varphi_i(t)} \end{aligned} \tag{2}$$

where $NLOS$ denotes the number of non-LOS paths.

The information implicit in multipath is significant but difficult to analyze. With deep learning, EUIGR takes the raw information as input directly, makes full use of information implied in the phase and RSS of the backscatter signal, and utilizes deep features to represent the gesture information.

Notations	Meaning
$C^{(t)}$	RFID Clip constructed by RFID data collector in time t
$F_R^{(t)}$	Feature representations extracted by the feature extractor in time t
$G_P^{(t)}$	Gesture probability vectors in time t
m	Length of the sliding time windows
n	Number of tags
$x_{k,tm}$	An element in RFID Clip which represents a phase and RSS pair $(phase, RSS)$ from the kth tag at the tm^{th} moment
θ_{fe}	The set of all parameters in the feature extractor
θ_{sl}	The set of all parameters in the LSTM sequence labeler
θ_{ud}	The set of all parameters of the user discriminator
θ_{ed}	The set of all parameters of the environment discriminator
NTtD	Normalized Time to Detect
UI	User Invariant
UIT	User Invariant Training
NT	Naive Training
EUI	Environment and User Invariant
EUIT	Environment and User Invariant Training

Table 1: Notations and Corresponding Meaning.

3.2 System Overview

As shown in Figure 1, the whole system consists of three main modules: RFID data collector, deep learning module and gesture predictor.

Before formally introducing the modules of EUIGR, for the sake of convenience, we concludes the notations in Table 1, which will be adopted later.

Firstly, RFID data collector collects and preprocesses the data streams of phase and RSS data from each RFID tag. Then preprocessed data streams of phase and RSS is divided into a set of sliding windows respectively, organized as RFID Clip $C^{(t)}$. The deep learning module consists of Multimodal CNN feature extractor, gesture sequence labeler, and two domain discriminators. Through adversarial learning, feature extractor maps the RFID Clip to environment independent, user invariant and gesture discriminative representations $F_R^{(t)}$, gesture sequence labeler models the temporal dynamics of representations stream and outputs a serial of gesture probability vectors $G_P^{(t)}$, and two domain discriminator plays a role of classifying the user who performs this gesture or positions where the gesture are executed. The deep learning module models the spatial structure and temporal dynamics of the spatio-temporal characters of gestures, generates the environment independent and user invariant representations, then calculates the probability of gestures in each time window. Finally, the gesture predictor module applies an SVM classifier which takes the features based on the

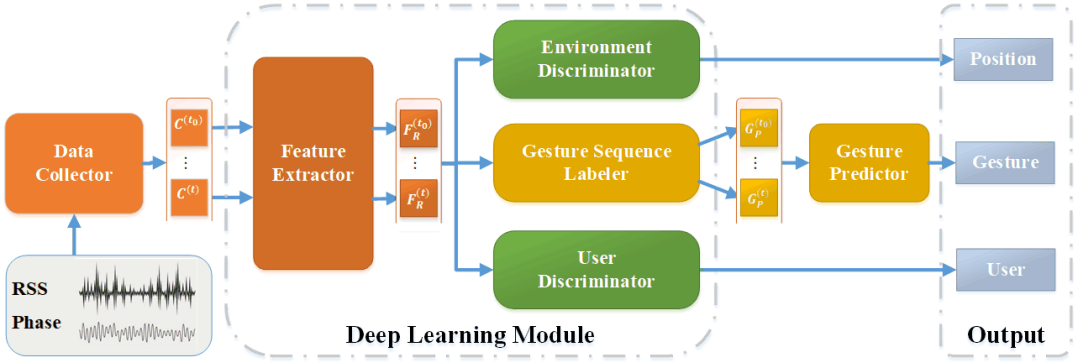


Figure 1: System Overview.

probabilities given by gesture sequence labeler as input to predict gestures with a negative delay.

4 SYSTEM METHODOLOGY

In this section, we show a detailed demonstration of the methodology of EUIGR and exhaustively analyze the core techniques, based on the aforementioned five basic functional components shown in Figure 1.

4.1 RFID Data Collector

RFID data collector is employed to collect and preprocess raw RSS and phase profiles of each RFID tag. The details of the preprocessing procedure are as follow:

Unwrapping Phase The raw phase profile of RFID signal reported by RFID reader is periodic multivalued, ranging from 0 to 2π which is also termed as a wrapped phase. This ambiguity of wrapped phase will increase the complexity of subsequent analysis. Fortunately, the tag reading rate is high enough and the speed of a human gesture is limited, so the difference between two adjacent phase values should be smaller than π [21]. Accordingly, we adopt an extensively applied phase unwrapping method[17].

Resampling In addition, the tag reply sequence is not uniform in the time domain[11], due to the random RFID tags respond to the reader, and packets loss caused by tag conflicts and noise. To cope with this problem, we resample the data stream at a frequency of $50Hz$. Moreover, before resampling, we employ the Hample identifier to filter out abnormal values in the stream.

Constructing RFID Clip After these operations, we divide the preprocessed stream into a set of sliding windows whose length is m (20 in EUIGR) with 50% overlap. Suppose that the reader with one antenna monitors n tags, such as $\{T_1, T_2, \ldots, T_n\}$. Then we convert all windows in the same period to an $n \times m$ matrix named RFID Clip , as follows:

$$C^{(tm)} = \begin{pmatrix} x_{1,t1} & x_{1,t2} & \cdots & x_{1,tm} \\ \vdots & \vdots & \vdots & \vdots \\ x_{k,t1} & x_{k,t2} & \cdots & x_{k,tm} \\ \vdots & \vdots & \vdots & \vdots \\ x_{n,t1} & x_{n,t2} & \cdots & x_{n,tm} \end{pmatrix}$$

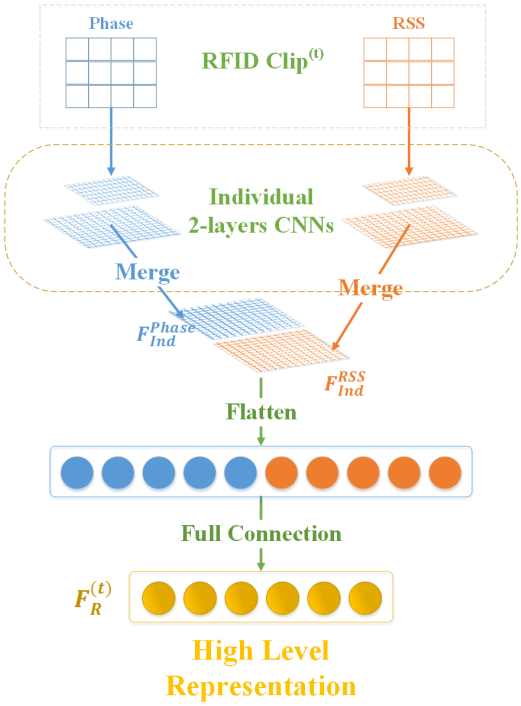


Figure 2: Feature Extractor.

where each row corresponds to a tag uniquely, while each column represents a description of the surrounding environment at a certain moment. Each element is a vector of phase and RSS data. To be specific, given an RFID Clip $C^{(tm)}$, tm is the ending time of this clip, and the element $x_{k,tm}$ represents phase and RSS pair $(phase, RSS)$ from the kth tag at the tm^{th} moment. Moreover, there will be a new RFID Clip generated every $200ms$ $(20 \div 50Hz \times 50\%)$ in EUIGR.

4.2 Feature Extractor

Human gesture recognition typically depends on data in different modalities, such as video, inertial sensing data, or low-level characters of wireless signals. Traditional methods usually manually

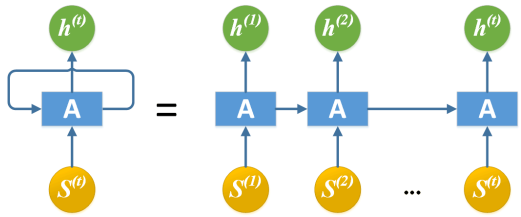


Figure 3: RNN Architecture. (The loop structure can unroll like chains which indicates that RNNs are intimately related to the sequence.)

select features from raw sensing inputs only in consideration of single modality, and then feed them into a classifier. However, an important challenge is to integrate seemingly incompatible sensor data which may immensely differ by its natural properties and data type. Motivated by the superior representation capability of CNNs, we leverage multimodal two-layer stacked CNNs to extract features from RFID Clip, which maximize the information of intra-modality and cross-modality.

Specifically, for each RFID Clip $C^{(t)}$, there are two $\mathbf{R}^{(n\times m)}$ data matrices corresponding to phase and RSS severally. We employ a 2-layer CNN on each individual data matrix, as revealed in Figure 2. The filter is a $(1 \times d)$ '2D' convolutional kernel so that we can extract intra-modality features. After convolutions, we merge the results F_{Ind}^{Phase} and F_{Ind}^{RSS} of these two subnets to get the output F_{Merge} of this layer. Eventually, after flattening, F_{Merge} is fed into a fully-connected layer to learn cross-modality information, and we can acquire the representation F_R of this RFID Clip as follows:

$$F_R^{(t)} = \mathbb{FE}(C^{(t)}; \theta_{fe}) \tag{3}$$

where θ_{fe} denotes the set of all parameters in the feature extractor $\mathbb{FE}$.

4.3 Gesture Sequence Labeler

We model the temporal dependencies of the RFID input sequences using RNN with LSTM. Thanks to that information can dynamically loop in RNN as exhibited in Figure 3, it addresses the problem caused by an arbitrarily long sequence which typically plagues other traditional architectures with a fixed number of input nodes. However, standard RNN has the problem of long-term dependencies due to gradient descent of an error criterion[2]. As an extended model of RNN, LSTM has a little more complex repeating structure instead of a single tanh layer of standard RNN[26]. To be more specific, the key of LSTM is its cell state, and LSTM has the ability to remove or add information to the cell state regulated by three gates. As a result, LSTM can remember information for long periods of time.

As shown in Figure 4, the cell state C_{t-1} and the hidden state h_{t-1} from the previous time clip, alongside the input x_t from the current clip are taken as input of a LSTM unit. In EUIGR, input x_t is the representations $F_R^{(t)}$ generated by feature extractor at clip t. The input gate i_t, forget gate f_t, update gate $\tilde{C}_t$ and output are calculated as:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{4}$$

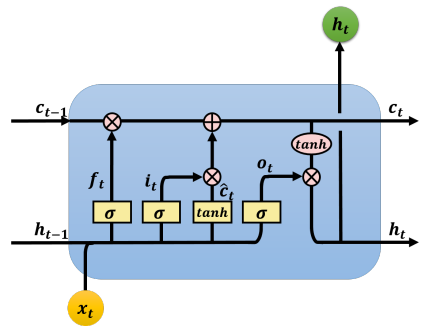


Figure 4: Structure of a LSTM cell.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{5}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{6}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{7}$$

where σ is the sigmoid function, and W and b stand for weight matrices and bias vectors, respectively. With these values, the LSTM unit updates C_t and h_t :

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \tag{8}$$

$$h_t = o_t * \tanh(C_t) \tag{9}$$

From these equations, we can see that LSTM can not only deal with the current input, but also make use of the previous context. Eventually, we project hidden state h_t into a low dimensional space and normalize it with softmax function, then interpret it as categorical probabilities vector of the gesture labels. The prediction probability of the kth gesture can be presented as:

$$G_P^{(t,k)} = \frac{\exp(h_t \cdot W_p + b_p)^k}{\sum_{k'} \exp(h_t \cdot W_p + b_p)^{k'}} \tag{10}$$

where W_p and b_p denote the weight matrix and bias vector of the projector.

In general, the $k \times 1$ prediction probability vector $G_P^{(t)}$ of all gestures can be represented as:

$$G_P^{(t)} = \mathbb{LSTM}_g(F_R^{(t)}; \theta_{sl}) \tag{11}$$

where θ_{sl} means a set of all parameters in LSTM sequence labeler. In general, the cross-entropy loss between the predicted gesture labels G_P and ground truths G_T is calculated as follows:

$$L_g(\theta_{fe}, \theta_{sl}) = -\frac{1}{N}\sum_{n=1}^{N}\frac{1}{T_n}\sum_{t=1}^{T_n}\sum_{k=1}^{K} G_T^{(t,k)} \log G_P^{(t,k)} \tag{12}$$

where N and K stand for the number of sequences and gesture categories, respectively, while the length of the nth sequence represented by T_n.

Different from some existing gesture recognition works, EUIGR takes not only all gestures but also *Non-gesture State* into consideration. *Non-gesture State* corresponds to all gestures or states which are not considered. Therefore, the number of gesture categories should be increased by one. In the simplest case, it can represent the gap between two consecutive gestures. Eventually, the sequence labeler outputs the probabilities of each gesture and *Non-gesture State* in every time clip.

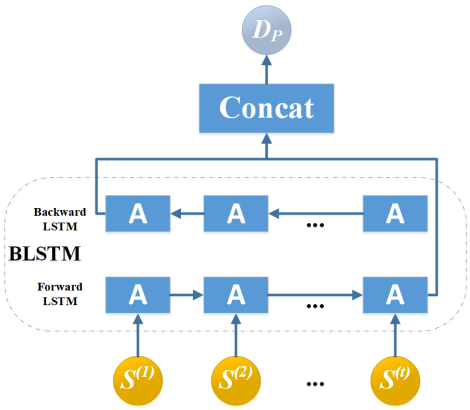


Figure 5: Structure of BLSTM.

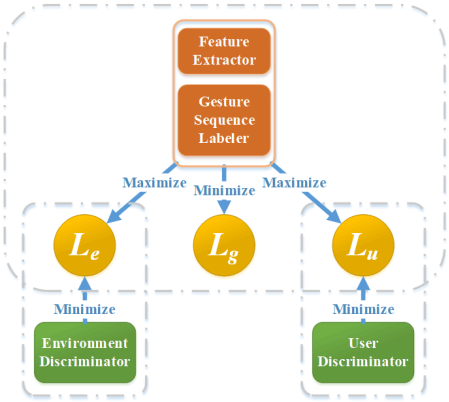


Figure 6: Adversarial training between two domain discriminators and gesture sequence labeler.

4.4 Domain Discriminators

In the EUIGR, we devote to modeling gestures by a more general and versatile feature representing method, which can be applied directly in unseen test users (none of their data are included in the training set) and untrained environments without adaptation. To put it from another angle, the feature generated by feature extractor should be as unrelated to the users and environments as possible. To achieve this goal, EUIGR further applies two domain discriminators including user discriminator and environment discriminator which maps the representations F_R to the user predictions and position predictions, respectively.

The data streams of the same gesture performed by diverse users in different positions may differ in both spatial and temporal. By intuition, the LSTM networks mentioned above can be utilized to modeling the temporal diversity among users or positions. Based on the details of LSTM elaborated in the previous subsection 4.3, it is obvious that LSTM is only able to make use of previous context and the output at time t is determined by the current input, as well as the previous information. Different from sequence labeler which predicts gesture labels in every time clip online, domain discriminators only work during the training process. As a consequence, we model the user diversity and environment discrepancy using Bidirectional LSTM(BLSTM) classifiers which contain two LSTM layers operating in opposite directions along the time domain as illustrated in Figure 5. Such a design ensures the inference about user or position depends on the full sequence with a better capability of temporal modeling.

At first, an user discriminator $\mathbb{BLSTM}_u$ is utilized to mining the probability distribution of users . Similar to equation (11), the user probability U_P about a certain sequence is revealed as:

$$U_P = \mathbb{BLSTM}_u(F_R; \theta_{ud}) \tag{13}$$

where θ_{ud} is the set of all parameters of user discriminator, and F_R denotes an isometric serial of representations depicted by feature extractor from the entire data sequence. In other words, unlike $\mathbb{LSTM}_g$ which deals with the representation of a single RFID Clip in equation (11), $\mathbb{BLSTM}_u$ takes the entire representations of a serial of RFID Clip as input and then gives a user prediction. At the same time, the cross-entropy loss L_u between the user predictions U_P and ground truths U_T can be presented as follows:

$$L_u(\theta_{fe}, \theta_{ud}) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{j=1}^{J} U_T^{(j)} \log U_P^{(j)} \tag{14}$$

where J means the number of user categories, and j is the index of user vectors. The goal of user discriminator is to minimize the above loss function so that the accuracy of user label predictions can be maximized.

Similar to the user discriminator $\mathbb{BLSTM}_u$, the environment discriminator is implemented by a BLSTM classifier $\mathbb{BLSTM}_e$, and the position inference E_P and the cross-entropy loss L_e between the position inference and ground truths E_T follow as:

$$E_P = \mathbb{BLSTM}_e(F_R; \theta_{ed}) \tag{15}$$

$$L_e(\theta_{fe}, \theta_{ed}) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{j=1}^{J} E_T^{(j)} \log E_P^{(j)} \tag{16}$$

, where the parameters resemble to those shown in Equation (13) and Equation (14).

4.5 Objective and Training Algorithm of deep learning module

As a matter of fact, directly optimizing the parameters θ_{fe} and θ_{sl} for minimizing loss L_g in Equation (12) usually succeeds on the users whose data are included in the training set in some certain positions in limited and controlled environment, but may get no effectiveness in unseen test users or unconsidered environment due to user variation and environment dependence.

To make the representation F_R gesture-discriminative, the distributions of the representations among different gestures should be as distinct to each other as possible. When it comes to environment-invariance or user-invariance, the features extracted from same kind of gestures performed by different users or in different positions are required to follow the same distribution. It is obvious that the purpose of two domain discriminators is opposite to our final objective. Therefore, all three losses are taken into consideration jointly, and the ultimate objective function is to minimize the total

Figure 7: Eight Traffic Command Gestures.

loss L_t:

$$L_t(\theta_{fe}, \theta_{sl}, \theta_{ud}, \theta_{ed}) = L_g(\theta_{fe}, \theta_{sl}) - \lambda_1 L_u(\theta_{fe}, \theta_{ud}) - \lambda_2 L_e(\theta_{fe}, \theta_{ed}) \quad (17)$$

where two coefficients λ_1 and λ_2 are devised for balancing the impact of user loss and environment loss in equation (14) and equation (16) severally, and both are set to a constant, 3.5 in EUIGR. As illustrated in Figure 6, three optimizers are employed to update all parameters of our model iteratively.

In the first optimizer, the parameters of feature extractor and sequence labeler are updated as follows:

$$(\theta_{fe}, \theta_{sl}) \leftarrow (\theta_{fe}, \theta_{sl}) - \alpha_1 \nabla_{(\theta_{fe}, \theta_{sl})} L_t(\theta_{fe}, \theta_{sl}, \theta_{ud}, \theta_{ed}) \quad (18)$$

where α_1 is the learning rate, and both θ_{ud} and θ_{ed} are considered as constants in gradient computing and will not be updated to keep the domain discriminator working.

In the other two domain optimizers, user discriminator is updated for keeping the ability to recognize the user domain of feature representations, while environment discriminator is for distinguishing the position area, go as follows:

$$\theta_{ud} \leftarrow \theta_{ud} - \alpha_2 \nabla_{\theta_{ud}} L_u(\theta_{fe}, \theta_{ud}) \quad (19)$$

$$\theta_{ed} \leftarrow \theta_{ed} - \alpha_3 \nabla_{\theta_{ed}} L_e(\theta_{fe}, \theta_{ed}) \quad (20)$$

where α_2 and α_3 mean the learning rate of two domain optimizers respectively, and θ_{fe} acts as constants in gradient calculation and will not be updated in domain optimizers. Through these opposite optimizations between the gesture recognizer and two domain discriminators, user-volatile and environment-specific factors are suppressed and features still keep gesture-representative.

4.6 Gesture Predictor

Through the sequence labeler, the gesture probability is given, and we can detect the occurrence of candidate gestures in a streaming fashion without the requirement of a separate and explicit segmentation step.

Traditionally, the system outputs the final prediction with gesture corresponding to the highest probability, if the probability is greater than the threshold, otherwise, the output is *Non-gesture State*. However, choosing an appropriate threshold will be a significant step in this method, and this chosen threshold may work well with some gestures but badly with others.

Accordingly, an SVM classifier based on probability values from gesture sequence labeler is employed as the gesture predictor instead of choosing an explicit threshold to detect gestures in advance. For incoming gesture label probability stream from the beginning (when the probability of the *Non-gesture State* is not the highest) to the tth time clip, we calculated the following features: 1) the normalized probability of each gesture at the tth time clip, excepts for *Non-gesture State*; 2) the mean of the normalized probabilities of each gesture individually from the beginning of this gesture to the tth time clip, excepts for *Non-gesture State*; 3) the variance of probabilities of each gesture (excepts for *Non-gesture State*) individually from the beginning of this gesture to the tth time clip, which describes the degree of volatility in each gesture predictions in the time domain; 4) the probability of *Non-gesture State* at the tth time clip; 5) At the same time, time interval from the beginning of this gesture to the tth time clip is also taken into consideration.

Feature	Term
Normalized Prob.	$\{NP^{(t,i)} = \frac{G_P^{(t,i)}}{\sum_{k=1}^{K} G_P^{(t,k)}}\}_{i=1}^{K}$
Mean of Prob.	$\{MP^{(i)} = \frac{1}{t-t_0} \sum_{j=t_0}^{t} NP^{(j,i)}\}_{i=1}^{K}$
Variance of Prob.	$\{VP^{(i)} = \frac{1}{t-t_0} \sum_{j=t_0}^{t} (NP^{(j,i)} - MP^{(i)})^2\}_{i=1}^{K}$
Non-gesture State	$G_P^{(t,K+1)}$
Time Interval	$t - t_0$

Table 2: Features used for gesture predictor.

As revealed in Table 2, all features are extracted from the gesture probabilities stream output by the gesture sequence labeler from the beginning time t_0 to current time t. Note that, K denotes the number of gestures while the $(K+1)^{th}$ index indicates *Non-gesture State*.

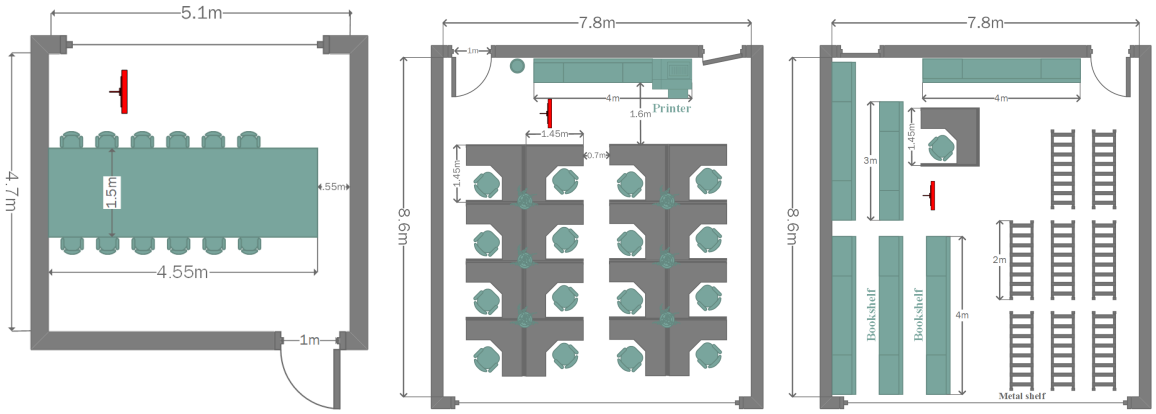
When the probability of *Non-gesture State* output by gesture sequence labeler is not the highest several times in a row, the gesture predictor is woken up and start to engendering the gesture prediction in every time clip. To the opposite, when the probability of *Non-gesture State* becomes the highest and it is held at several subsequent time clips, the gesture predictor enters the sleep stage again.

5 EXPERIMENTAL RESULTS

5.1 Implementation and Deployment

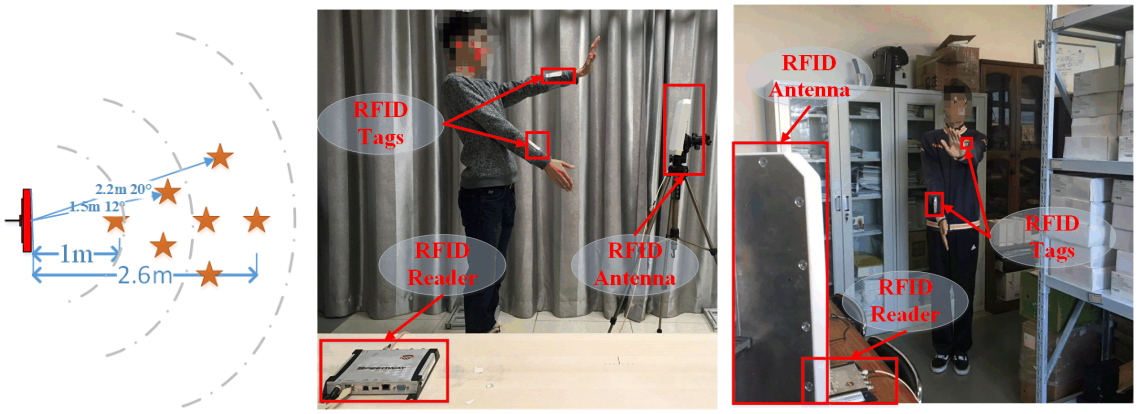
Implementation: Our implementations are purely based on the COTS devices without any hardware modifications.

We employ Impinj Speedway R420 RFID reader equipped with a Laird S9028PCR antenna ($9dBi$). The reader operates at $920MHz$ with a transmitting power of $24dBm$ and sensitivity of $-80dBm$. The



(a) Meeting Room (b) Office (c) Warehouse

Figure 8: Floor plans of our three experimental rooms. Red rectangles stand for the antennas.



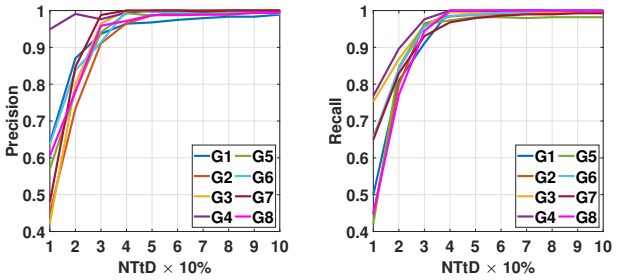
(a) Seven Relative Positions (b) Deployment in Meeting Room (c) Deployment in Warehouse

Figure 9: Deployments.

type of tags is ALN-9762 ($70mm \times 17mm$). And we train and optimize our model on a customized Linux server with an Intel(R) Core(TM) i9-7900X 3.30GHz 10-core CPU and 4 GeForce GTX 1080Ti GPUs. The runtime evaluation of EUIGR is conducted on a laptop (Lenovo Y50-70) equipped with an Intel(R) Core(TM) i5-4200H 2.80GHz dual-core CPU and a GeForce GTX 860m GPU. The signal collector is implemented using C# with LLRP[7]. Other modules are realized on the Python 3.6 platform, where the deep learning module is constructed through Tensorflow. In the deep learning module, the number of hidden cells in feature extractor corresponding to 2-layer CNNs are 64 and 64, respectively, and the fully connected layer is 128. Both user discriminator and environment discriminator are built by a one-layer BLSTM where two opposite LSTM layers contain 32 hidden cells individually, while each layer in LSTM of gesture sequence labeler contains 64 hidden cells.

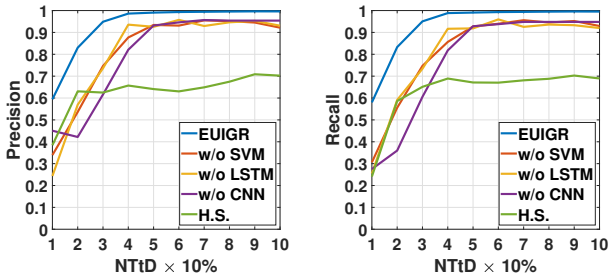
Dataset: Fifteen volunteers are invited to participate in our experiments. They are 12 male and 3 female university students with the age ranging from 21 to 30. The gestures performed by volunteers are eight traffic command gestures which are adopted by traffic police to guide vehicles in some regions. As shown in Figure 7, these gestures include turn right, turn left, left turn pending, go straight, pull over, stop, slow down and change lane. Each volunteer is required to perform each gesture at least 15 times repeatedly in random order. Volunteers are not required to stand in the exact positions or face the antenna strictly.

Besides, all 7 possible relative positions of the volunteers standing area and the antenna are in the interrogation area of the antenna. These relative positions are illustrated in Figure 9(a), where orange stars denote the possible areas where users are required to stand on, blue line means distance between the antenna and these positions, red rectangle stands for the antenna, and gray dotted lines represent the interrogation area. They may repeatedly appear in three distinct rooms including a meeting room, an office with more than a dozen people and PCs, and a warehouse full of goods. The floor plans of three rooms are illustrated in Figure 8, and the warehouse contains multiple metal shelves filled with electronic instruments or



(a) Precision (b) Recall

Figure 10: General performance (w.r.t. NTtD) of EUIGR.



(a) Precision (b) Recall

Figure 11: Comparison with other competitive methods.

books as shown in Figure 9(c). Complex environments may induce little variance on the system inputs, but the changes of the relative positions between the users and the antenna will induce obvious variance on the received phases and RSS. Though four of seven relative positions appear in at least two rooms, the other three only appear once. The same relative positions in different rooms are seen as different positions in the environment domain, then 12 different categories in the environment domain are included in the experiment eventually. The entire gesture samples collection work last for three weeks and more than five thousand samples are obtained. As depicted in Figure 9, a volunteer stands about 1.0 to 2.6 m in front of the RFID antenna and wears a tag on each wrist when they are performing the gestures. In addition, the samples in user-mixture model and the samples in user independent evaluation are collected from the gestures performed by 15 users in the position shown in Figure 9(b) in more than dozen days.

Evaluation Metric: Our principal goal is to evaluate the capability of ongoing gesture recognition, so we compute the precision and recall of our model as a function of different fraction of the total gesture duration. More precisely, we compute the gesture detection accuracy for various values of *Normalized Time To Detect (NTtD)*[15] (in increments of 10%), which is the portion of gesture that has been completed.

5.2 General Performance of EUIGR

We first evaluate the general performance of the user-mixture mode where each gesture instance is a primitive unit and may come from any volunteer. The evaluation results are averaged by 10-fold cross-validation. Figure 10 depicts the precision and recall for eight gestures as a function of the NTtD value. We can observe that when the NTtD value is close to 50%, the precision and recall of any gesture are higher than 96.8% and 98.0% severally, and average precision and recall are higher than 99.0%. As the NTtD value continues to increase, the average precision and recall are eventually higher than 99.5%.

5.3 Comparison with Other Methods

To further demonstrate the superiority of our elaborate algorithm, we compare EUIGR with other competitive methods, which are some representative methods or the variants of our model by removing one component in the architecture,

w/o SVM: We remove the SVM predictor to study the utility of the gesture predictor. In this case, we simply take the gesture with the highest probability as the output.

w/o LSTM: There are no LSTM component in this model, and we take the output of multimodal CNN as the label prediction probabilities of each time clip in both model training and testing procedure. We evaluate this model for depicting the significance of longtime dependency in gesture recognition.

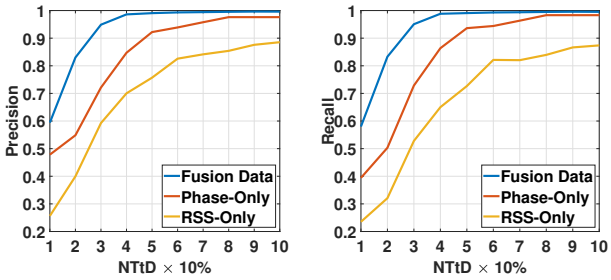
w/o CNN: In this model, we directly feed the RFID clip into LSTM without CNN feature extraction. We evaluate this model to show the importance of CNN in discovering deep features.

H.S.: We adopt a Hidden Markov Model (HMM) and SVM (Supported Vector Machine) based method in [33]. Feature Extractor in EUIGR is replaced by a K-Means encoder, a Left-To-Right HMM [33] which plays the role of Gesture Sequence Labeler, while Gesture Predictor is still an SVM classifier. The role of this model is to highlight the effectiveness of the deep learning module.

Model	EUIGR	w/o SVM	w/o LSTM	w/o CNN	H.S.	GRfid
Precision	0.996	0.921	0.930	0.954	0.702	0.630
Recall	0.995	0.929	0.920	0.948	0.689	0.608

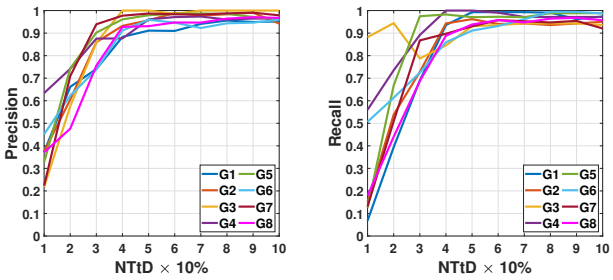
Table 3: Comparison of final performance between EUIGR and other competitive methods.

Firstly, the final performance of the completed gestures is evaluated and the results are shown in Table 3, where the performance of the method in GRfid [44] is evaluated on the traffic command gestures. Both precision and recall values with regard to NTtD are illustrated in Figure 11 afterward. Apparently, EUIGR achieves better performance than other variant models. Moreover, traditional machine learning methods perform much worse than the deep learning methods. Traditional machine learning methods have been widely used in the gesture recognition field, and many of the gestures studied are periodic and have a short duration. On the contrary, the traffic command gestures which involve a sequence of actions have longer performing durations and larger motion amplitudes. Hence it is not intuitive to find the corresponding signal patterns, which makes extracting and matching the gesture patterns for gesture recognition more difficult and troublesome.



(a) Precision (b) Recall

Figure 12: Comparison between model with both RSS and phase fusion data, phase-only data, and RSS-only data.



(a) Precision (b) Recall

Figure 13: General performance of UIT.

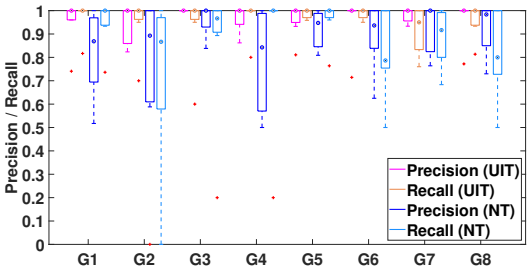


Figure 14: Comparison between UIT and NT.

This is why those traditional machine learning methods perform poorly. Compared with all of these competitive methods, EUIGR makes more efficient use of long-term temporal relations, extracts more comprehensive deep features, and generates more accurate predictions. These comparative results reinforce the necessity of our design components in EUIGR.

5.4 Significance of Multimodal Data Fusion

In this subsection, we discuss the importance of multimodal RFID data fusion. EUIGR applies a CNN based data fusion technique to integrate RSS and phase information. We evaluate our model by feed different inputs, including fusion data, phase-only data, and RSS-only data on the user-mixture mode. The comparison results are provided in Figure 12. Evidently, the information carried by phase is more detailed than RSS in representing gestures, due to all inputs with phase data can enable the model getting a high final recognition rate (after the gestures are completed). However, when both phase and RSS are taken as input jointly, our model achieves higher precision and recall in the early gesture ongoing stage, while the phase-only input leads to poor performance. This is because the phase and RSS carry information on different respects. In other words, in the early stage of gesture execution, we can only obtain a small amount of information about the existing phase or RSS stream. However, the richer the information, the more accurate the recognition. In the later stage, the information contained in the phase is sufficient, so the phase-only data can also result in a high final performance.

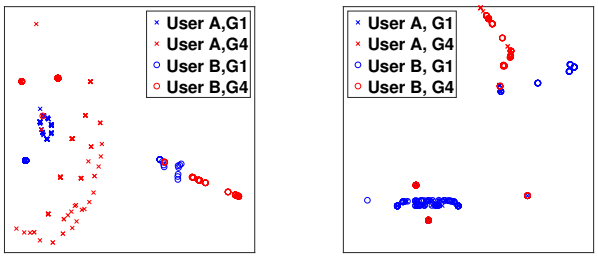
5.5 Performance of UI Model

Evaluations above show that EUIGR achieves great success on the dataset of known users, and we will discuss the performance in unseen users and demonstrate the significance of adversarial architecture in this subsection. We utilize the Leave One Participant Out (LOPO) cross-validation technique, whereby we train the User Invariant (UI) model using the samples from $J-1$ volunteers and test the system using the samples from another volunteer, repeating this process for every distinct volunteer.

As shown in Figure 13, the precision and recall over eight gestures are measured as a function of the NTtD with the UI model. As expected, the person-independent model gives rise to lower accuracy than the user-mixture model but also achieves high performance. Both the precision and recall of all gestures are higher than 91.1%, and the average of precision and recall are higher than 96.1% and 95.4% severally when the NTtD value is close to 50%.

To show the progress in the UI model more intuitively, we compare the final gesture recognition rate (after the gestures are completed) between the model with User Invariant Training (UIT) via adversarial learning and Naive Training (NT) model without user discriminator. The comparison results are presented in Figure 14. On each box, the central mark expresses the median, and the bottom and top edges of the box represent the 25th and 75th percentiles, respectively. The whiskers extend to the most extreme data points (outliers are not considered), and the outliers are depicted individually using the '+' symbol. For each gesture, we compare the precision and recall of UIT and NT severally, and we conclude that both precision and recall performance about UIT are more stable and high than NT across users, due to user-variant is eliminated by UIT. UIT leads to the average 96.6% of precision and 96.0% of recall compared with the 87.2% and 86.2% of NT respectively.

To explore the reason why we get better performance with adversarial learning, we visualize the representations F_R extracted by the feature extractor using t-SNE. We randomly select two kinds of gestures performed by two volunteers and extract the latent representations F_R corresponding the median clip of these gestures. The representations generated on the NT and UIT are illustrated in Figures 15 by t-SNE. In Figure 15(a), we can obviously observe that the distributions of both different users and gestures are separated. The distributions of the same gesture performed by different users



(a) NT (b) UIT

Figure 15: Representations output from Feature Extractor.

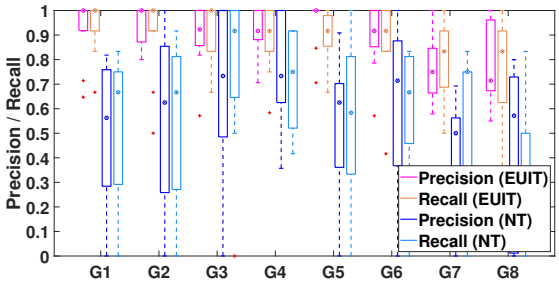


Figure 16: Comparison between EUIT and NT.

are even more discrete than the different gesture performed by the same users. With UIT, the latent feature distributions of the same gestures performed by different participants are overlapped and cannot be distinguished, as revealed in Figure 15(b). In addition, the feature distributions from different gestures are still far away from each other, and we can say that these features are discriminating in gestures and independent among users. In conclusion, with adversarial learning, the feature extractor possesses the remarkable ability to generate user-invariant features, and our model achieves high performance in LOPO cross-validation.

5.6 Performance of EUI Model

In previous subsection, the user invariant training is evaluated and the great success is obtained in the user independent model. At the same time, the reason why adversarial learning brings high performance is elaborated and demonstrated by feature visualization. So, naturally, it comes to the environment-invariance of our approach. In Environment and User Invariant (EUI) evaluation, all samples are collected from the gestures performed by several users in 12 positions in three weeks. Similar to the evaluation method of UI model, the LOPO cross-validation technique is utilized in EUI model with a little adaption where the participants here mean not different users but positions and surrounding environments. To be more specific, the samples of two users from a same position are in different domains among users but in the same domain among environments, and are regarded as performed by the same 'participant', will only take part in the training process or testing process. Firstly, the comparison results of final precision and recall of gesture recognition between EUIT and NT are shown in Figure 16. EUIT leads to the average 88.6% of precision and 86.7% of recall which is much higher than the average about 56% of precision and recall caused by NT.

A detailed comparison between Figure 14 and Figure 16 reveals that the final recognition performance of gesture G7 and G8 attenuates severely. These two gestures themselves reveal the reason. First of all, both G7 and G8 require users swing only right hand twice, while the difference between them is just the swing direction of their right hand. The swing direction of users' right hand can be transfered into the relative moving direction of the RFID tag attached on right hand and the antenna. When the user independent performance is evaluated, all users are required to stand in the

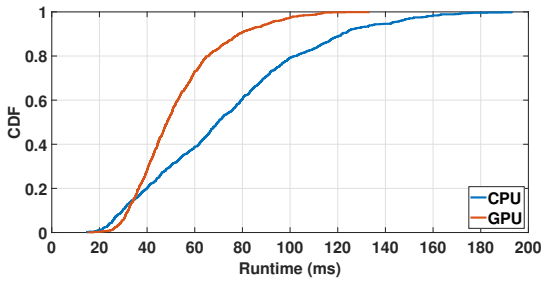


Figure 17: Runtime of EUIGR.

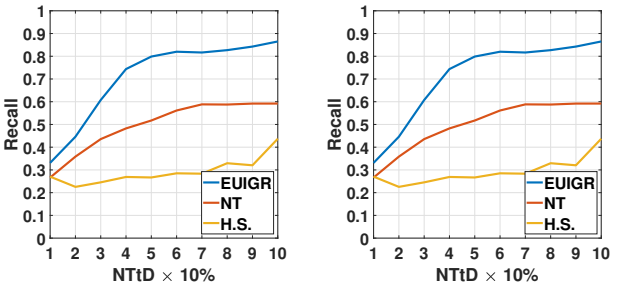
position in front of the antenna, both G7 and G8 have their own relative moving direction patterns. However, when it comes to the position changeable experiments, users may turn around slightly or move to another side of the interrogation area of the antenna, this relative moving direction patterns of G7 and G8 become similar to each other, leading the confusion between these two gestures.

5.7 Real Time Performance

The previous evaluations have demonstrated that our approach can recognize gestures early and accurately (with recall of 98% for the user-mixture model and recall of 96% for the user-independent model respectively at the midpoint of a gesture). At the same time, we evaluate the runtime of EUIGR by predicting 1000 samples in a row using the CPU and GPU on the laptop respectively, the *Cumulative Distribution Function* (CDF) of time cost is illustrated in Figure 17. We can conclude that even with the CPU, nearly 50% time costs are less than 70*ms* and almost all time costs are less than 180*ms*. The traffic command gestures generally last 4 to 6 seconds, while 180*ms* accounts for less than 5% of this time segment. Take the length of the time window (as mentioned in Subsection 4.1, 200*ms*) into consideration, we can output a gesture prediction before next time window comes.

6 DISCUSSION

In this section, we discuss several final thoughts on the future of EUIGR.



(a) Precision (b) Recall

Figure 18: General performance of eight alphabet gestures.

In this paper, we just take eight traffic command gestures to demonstrate the effectiveness and accuracy of EUIGR. The proposed method can be applied to other wrist-worn-tag gestures. Actually, we have collected eight gestures of writing letters of the alphabet (A to H) and evaluated the proposed method on these gestures for demonstrating the effectiveness and generality of EUIGR. The deployments of alphabet writing gestures set are similar to traffic command gestures. To be specific, a volunteer whose right wrist is attached an RFID tag are required to perform each gesture at least 15 times repeatedly in random order. The seven relative positions keep unchanged as shown in Figure 9(a) but the surrounding environments have some changes where the original meeting room is replaced by an NLoS scenario. In the NLoS scenario, the antenna and the user are in two rooms and separated by a wood siding wall. Eventually, there are three different environments and seven relative positions, and three of seven relative positions have only appeared once, then thirteen different positions are included in the experiment. Compared with traffic command gestures, alphabet writing gestures with a duration of 2.5 to 3.5 seconds have a shorter duration and a slighter motion amplitude. The comparative evaluation with other competitive models is carried out on the alphabet writing gesture set. Both the average precision and recall values over eight alphabet writing gestures as a function of NTtD is depicted in Figure 18. On the alphabet writing gestures, EUIGR also obtained an average precision and recall of higher than 85% in untrained environments. Moreover, EUIGR performs much better than the NT model and H.S. model, which validates the effectiveness and generality of EUIGR. Besides, collecting the rest of the alphabet writing gestures and completing the gesture set is left as our future work.

In addition, richer signal information or customized service usually provides higher performance. The hardware setup in EUIGR is not complex and the effectiveness of its design is validated illustratively. The trade-off between the performance and the simplicity of deployment is always an interesting topic, we will leave it as our future work. Though about 86% of the performance of EUIGR with untrained positions may not be satisfactory in some applications, it can easily be promoted by transfer learning with fine-tuning to adapt to a certain environment.

7 CONCLUSIONS

In this paper, we elaborate EUIGR, an RFID based environment invariant and user independent real-time gesture recognizing system. EUIGR integrates multimodal CNN and LSTM seamlessly well to represent features in RFID signals. With adversarial learning, EUIGR suppresses the effect of the user-volatile and environment-specific factors in the traditional model. To be more specific, the gesture sequence labeler and user discriminator are jointly optimized to maximize the user and position classification loss and minimize the gesture labeling loss, via EUIT. Through this multi-task optimization, the feature extractor is trained to generate the user-invariant, environment-independent, and gesture-discriminative representations. To support natural gestural interaction in immersive applications, EUIGR adopts an SVM classifier that takes the features based on the probabilities output by gesture sequence labeler as input to detect ongoing gestures. We implement and deploy EUIGR in a wrist-worn-tag scenario with COTS RFID devices to evaluate its performance. Extensive experimental results show that EUIGR achieves an average recognition accuracy of 96% even at the midpoint of traffic command gestures on unseen users. In untrained positions or rooms, EUIGR obtains the average 88.6% of precision and 86.7% recall in final recognition. The encouraging performance demonstrates that EUIGR provides significant high accuracy, low latency, and strong robustness. We believe that the proposed adversarial architecture is compatible with other tasks, and our future work will focus on extracting user-specific and action-invariant features to enable user identification and safety certification.

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